ChatGPT

Yes. In fact the code makes $a_2=2f(n)+½$ slightly above 1 (since $f(n)\to¼$ as $n\to\infty$), so the faster therapy is the second one. The constraint $x_1 + a_2x_2 =1$ is thus minimized by putting $x_1=0$ and $x_2=1/a_2\approx1$. The infinity-norm distance from $(0.1,0.9)$ to the point $(0,1)\approx(0,1/a_2)$ is $\max(|0.1-0|,\;|0.9-1|)=0.1$, which is ≤0.15. Hence $(0.1,0.9)$ is at most 0.15 away from a minimiser.
